

```
# =====  
#      STEP 8 – BUSINESS INSIGHTS & RECOMMENDATIONS  
# =====  
# Goal:  
# Convert the analysis results into actionable business  
# insights and exactly 5 business recommendations.  
#  
# Outputs:  
# 1. Business KPI summary  
# 2. Customer insights  
# 3. Product insights  
# 4. Purchase behavior insights  
# 5. Churn insights  
# 6. RFM segment insights  
# 7. Exactly 5 actionable recommendations  
# 8. Business recommendations CSV  
# 9. Executive summary TXT  
#  
# =====  
  
# -----  
# STEP 1 – IMPORT LIBRARIES  
# -----
```

```
import pandas as pd  
import numpy as np  
from pathlib import Path
```

```
# -----  
# STEP 2 - CREATE OUTPUT FOLDER  
# -----
```

```
output_folder = Path("step8_business_insights")
```

```
output_folder.mkdir(exist_ok=True)
```

```
# =====  
# STEP 3 - LOAD PHASE 7 DATA  
# =====
```

```
base_folder = Path("step7_powerbi_data")
```

```
fact_sales = pd.read_csv(  
    base_folder / "fact_sales.csv"  
)
```

```
customer_kpis = pd.read_csv(  
    base_folder / "customer_kpis.csv"  
)
```

```
monthly_sales = pd.read_csv(  
    base_folder / "monthly_sales.csv"  
)
```

```
category_performance = pd.read_csv(|
```

```
    base_folder / "category_performance.csv"
)

payment_performance = pd.read_csv(
    base_folder / "payment_performance.csv"
)

day_performance = pd.read_csv(
    base_folder / "day_performance.csv"
)

hour_performance = pd.read_csv(
    base_folder / "hour_performance.csv"
)

time_period_performance = pd.read_csv(
    base_folder / "time_period_performance.csv"
)

rfm_segment_summary = pd.read_csv(
    base_folder / "rfm_segment_summary.csv"
)

segment_churn = pd.read_csv(
    base_folder / "segment_churn.csv"
)
```

|

```
# -----
# STEP 4 - LOAD RFM DATA
# -----

rfm = pd.read_csv(
    "customer_rfm_segments.csv"
)

# =====
# STEP 5 - CONVERT NUMERIC COLUMNS
# =====

numeric_columns = [
    "total_purchase_amount",
    "quantity",
    "product_price",
    "Revenue"
]

for col in numeric_columns:

    if col in fact_sales.columns:

        fact_sales[col] = pd.to_numeric(
            fact_sales[col],
            errors="coerce"
        )
```

```
#####  
# SECTION A - OVERALL BUSINESS PERFORMANCE  
# =====  
  
# -----  
# STEP 6 - CALCULATE MAIN KPIs  
# -----  
  
total_revenue = fact_sales[  
    "total_purchase_amount"  
].sum()  
  
total_transactions = len(fact_sales)  
  
total_customers = fact_sales[  
    "customer_id"  
].nunique()  
  
average_order_value = (  
    total_revenue /  
    total_transactions  
)  
  
total_quantity = fact_sales[  
    "quantity"  
].sum()  
  
average_quantity = (  
    total_quantity /
```

```
total_transactions
)

# -----
# STEP 7 - RETURN KPI
# -----

if "Return_Flag" in fact_sales.columns:

    total_returns = fact_sales[
        "Return_Flag"
    ].sum()

    return_rate = (
        total_returns /
        total_transactions
    ) * 100

else:

    total_returns = 0
    return_rate = 0
```

```
# -----  
# STEP 8 – CHURN KPI  
# -----
```

```
if "churn" in customer_kpis.columns:
```

```
    churned_customers = (  
        customer_kpis["churn"]  
        .fillna(0)  
        .sum()  
    )
```

```
    churn_rate = (  
        churned_customers /  
        total_customers  
    ) * 100
```

```
else:
```

```
    churned_customers = 0  
    churn_rate = 0
```

```
# -----  
# STEP 9 – DISPLAY OVERALL KPIS  
# -----
```

```
business_kpis = pd.DataFrame({
```



```

business_kpis = pd.DataFrame({
    "Metric": [
        "Total Customers",
        "Total Transactions",
        "Total Revenue",
        "Average Order Value",
        "Total Quantity",
        "Average Quantity per Transaction",
        "Total Returns",
        "Return Rate (%)",
        "Churned Customers",
        "Churn Rate (%)"
    ],
    "Value": [
        total_customers,
        total_transactions,
        total_revenue,
        average_order_value,
        total_quantity,
        average_quantity,
        total_returns,
        return_rate,
        churned_customers,
        churn_rate
    ]
})

print("\n===== BUSINESS KPIs =====\n")

```



```
display(business_kpis)
```

```
# =====
```

```
# SECTION B – PRODUCT INSIGHTS
```

```
# =====
```

```
# -----
```

```
# STEP 10 – BEST CATEGORY BY REVENUE
```

```
# -----
```

```
best_category_row = category_performance.loc[  
    category_performance["Revenue"].idxmax()  
]
```

```
best_category = (  
    best_category_row["product_category"]  
)
```

```
best_category_revenue = (  
    best_category_row["Revenue"]  
)
```

```
# -----  
# STEP 11 – MOST TRANSACTED CATEGORY  
# -----  
  
most_purchased_category_row = (  
    category_performance.loc[  
        category_performance["Transactions"].idxmax()  
    ]  
)  
  
most_purchased_category = (  
    most_purchased_category_row[  
        "product_category"  
    ]  
)  
  
# -----  
# STEP 12 – CATEGORY WITH HIGHEST RETURN RATE  
# -----  
  
if "Return_Rate" in category_performance.columns:  
  
    highest_return_category_row = (  
        category_performance.loc[  
            category_performance["Return_Rate"].idxmax()  
        ]  
    )  
  
|
```

```
highest_return_category = (  
    highest_return_category_row[  
        "product_category"  
    ]  
)
```

```
highest_return_rate = (  
    highest_return_category_row[  
        "Return_Rate"  
    ]  
)
```

else:

```
highest_return_category = "N/A"  
highest_return_rate = 0
```

```
# -----  
# STEP 13 - CREATE PRODUCT INSIGHT TABLE  
# -----
```

```
product_insights = pd.DataFrame({  
    "Insight": [  
        "Highest Revenue Category",  
        "Most Purchased Category",  
        "Highest Return Rate Category"  
    ],  
    |
```

```

    "Value": [
        best_category,
        most_purchased_category,
        highest_return_category
    ],
    "Metric": [
        round(best_category_revenue, 2),
        int(
            most_purchased_category_row[
                "Transactions"
            ]
        ),
        round(highest_return_rate, 2)
    ]
})

print("\n===== PRODUCT INSIGHTS =====\n")

display(product_insights)

```

```
# =====  
# SECTION C – PURCHASE BEHAVIOR INSIGHTS  
# =====  
  
# -----  
# STEP 14 – BEST DAY BY TRANSACTIONS  
# -----  
  
best_day_row = day_performance.loc[  
    day_performance["Transactions"].idxmax()  
]  
  
best_purchase_day = (  
    best_day_row["Day_Name"]  
)  
  
best_day_transactions = (  
    best_day_row["Transactions"]  
)  
  
# -----  
# STEP 15 – BEST HOUR BY TRANSACTIONS  
# -----  
  
best_hour_row = hour_performance.loc[  
    hour_performance["Transactions"].idxmax()  
]
```

```
best_purchase_hour = (  
    best_hour_row["Purchase_Hour"]  
)  
  
best_hour_transactions = (  
    best_hour_row["Transactions"]  
)  
  
# -----  
# STEP 16 - BEST TIME PERIOD  
# -----  
  
best_time_row = time_period_performance.loc[  
    time_period_performance["Transactions"].idxmax()  
]  
  
best_time_period = (  
    best_time_row["Time_Period"]  
)  
  
best_time_transactions = (  
    best_time_row["Transactions"]  
)  
  
#
```

```
# -----  
# STEP 17 – BEST MONTH BY REVENUE  
# -----
```

```
best_month_row = monthly_sales.loc[  
    monthly_sales["Revenue"].idxmax()  
]
```

```
best_month = (  
    best_month_row["Month_Year"]  
)
```

```
best_month_revenue = (  
    best_month_row["Revenue"]  
)
```

```
# -----  
# STEP 18 – CREATE PURCHASE INSIGHT TABLE  
# -----
```

```
purchase_insights = pd.DataFrame({  
    "Insight": [  
        "Best Purchase Day",  
        "Peak Purchase Hour",  
        "Best Time Period",  
        "Highest Revenue Month"  
    ],  
    },
```



```

        "Value": [
            best_purchase_day,
            str(int(best_purchase_hour)) + ":00",
            best_time_period,
            best_month
        ],
        "Metric": [
            best_day_transactions,
            best_hour_transactions,
            best_time_transactions,
            best_month_revenue
        ]
    })

print("\n===== PURCHASE BEHAVIOR INSIGHTS =====\n")

display(purchase_insights)

```

```

# =====
# SECTION D – CUSTOMER VALUE INSIGHTS
# =====

# -----
# STEP 19 – TOP CUSTOMER
# -----

if "Total_Spend" in customer_kpis.columns:

```

```
if "Total_Spend" in customer_kpis.columns:

    top_customer_row = customer_kpis.loc[
        customer_kpis["Total_Spend"].idxmax()
    ]

    top_customer_id = (
        top_customer_row["customer_id"]
    )

    top_customer_spend = (
        top_customer_row["Total_Spend"]
    )

else:

    top_customer_id = "N/A"
    top_customer_spend = 0
```

```
# -----
# STEP 20 - AVERAGE CUSTOMER SPEND
# -----
```

```
if "Total_Spend" in customer_kpis.columns:

    average_customer_spend = (
        customer_kpis["Total_Spend"]
```

```
        .mean()
    )

else:

    average_customer_spend = 0

# -----
# STEP 21 – AVERAGE CUSTOMER FREQUENCY
# -----

if "Purchase_Frequency" in customer_kpis.columns:

    average_customer_frequency = (
        customer_kpis["Purchase_Frequency"]
        .mean()
    )

else:

    average_customer_frequency = 0
```

```
# -----  
# STEP 22 - CREATE CUSTOMER INSIGHT TABLE
```

```
# -----
```

```
customer_insights = pd.DataFrame({  
    "Insight": [  
        "Top Customer",  
        "Top Customer Spend",  
        "Average Customer Spend",  
        "Average Purchase Frequency"  
    ],  
    "Value": [  
        top_customer_id,  
        top_customer_spend,  
        average_customer_spend,  
        average_customer_frequency  
    ]  
})
```

```
print("\n===== CUSTOMER VALUE INSIGHTS =====\n")
```

```
display(customer_insights)
```

```
# =====
# SECTION E – RFM SEGMENT INSIGHTS
# =====

# -----
# STEP 23 – LARGEST CUSTOMER SEGMENT
# -----

largest_segment_row = rfm_segment_summary.loc[
    rfm_segment_summary["Customers"].idxmax()
]

largest_segment = (
    largest_segment_row["Customer_Segment"]
)

largest_segment_customers = (
    largest_segment_row["Customers"]
)

# -----
# STEP 24 – HIGHEST REVENUE SEGMENT
# -----

highest_revenue_segment_row = (
    rfm_segment_summary.loc[
        rfm_segment_summary["Revenue"].idxmax()
    ]
)
```

```
)  
  
highest_revenue_segment = (  
    highest_revenue_segment_row[  
        "Customer_Segment"  
    ]  
)
```

```
highest_revenue_segment_value = (  
    highest_revenue_segment_row[  
        "Revenue"  
    ]  
)
```

```
# -----  
# STEP 25 - HIGHEST CHURN SEGMENT  
# -----
```

```
if "Churn_Rate" in segment_churn.columns:  
  
    highest_churn_segment_row = (  
        segment_churn.loc[  
            segment_churn["Churn_Rate"].idxmax()  
        ]  
    )
```

```
| highest_churn_segment = (  
    highest_churn_segment_row[  
        "Customer_Segment"  
    ]  
)
```

```
        highest_churn_segment_row[
            "Customer_Segment"
        ]
    )
```

```
highest_churn_rate = (
    highest_churn_segment_row[
        "Churn_Rate"
    ]
)
```

else:

```
highest_churn_segment = "N/A"
highest_churn_rate = 0
```

```
# -----
# STEP 26 - CREATE RFM INSIGHT TABLE
# -----
```

```
rfm_insights = pd.DataFrame({
    "Insight": [
        "Largest Customer Segment",
        "Highest Revenue Segment",
        "Highest Churn Segment"
    ],
```

```

    "Value": [
        largest_segment,
        highest_revenue_segment,
        highest_churn_segment
    ],
    "Metric": [
        largest_segment_customers,
        highest_revenue_segment_value,
        highest_churn_rate
    ]
})

```

```

print("\n===== RFM INSIGHTS =====\n")

```

```

display(rfm_insights)

```

```

# =====
# SECTION F – CHURN INSIGHTS
# =====

```

```

# -----
# STEP 27 – CHURNED REVENUE
# -----

```

```

if {
    "churn",
    "Total_Spend"
}

```



```
}.issubset(customer_kpis.columns):
```

```
    churned_revenue = (  
        customer_kpis.loc[  
            customer_kpis["churn"] == 1,  
            "Total_Spend"  
        ].sum()  
    )
```

```
    active_revenue = (  
        customer_kpis.loc[  
            customer_kpis["churn"] == 0,  
            "Total_Spend"  
        ].sum()  
    )
```

```
else:
```

```
    churned_revenue = 0  
    active_revenue = 0
```

```
# -----  
# STEP 28 - CHURNED REVENUE PERCENTAGE  
# -----
```

```
total_customer_revenue = (  
    churned_revenue +
```

```
    active_revenue
```

```
)
```

```
if total_customer_revenue > 0:
```

```
    churned_revenue_percentage = (  
        churned_revenue /  
        total_customer_revenue  
    ) * 100
```

```
else:
```

```
    churned_revenue_percentage = 0
```

```
# -----
```

```
# STEP 29 - HIGH-VALUE AT-RISK CUSTOMERS
```

```
# -----
```

```
if {
```

```
    "R_Score",
```

```
    "M_Score",
```

```
    "churn"
```

```
}.issubset(rfm.columns):
```

```
    high_value_at_risk = rfm[
```

```
        (
```

```
            (rfm["R_Score"] <= 2) &
```

```
        (rfm["M_Score"] >= 4) &
        (rfm["churn"] == 0)
    )
].copy()
```

else:

```
    high_value_at_risk = pd.DataFrame()
```

```
high_value_at_risk_count = len(
    high_value_at_risk
)
```

```
# -----
# STEP 30 - CHURN INSIGHT TABLE
# -----
```

```
churn_insights = pd.DataFrame({
    "Insight": [
        "Total Churned Customers",
        "Overall Churn Rate (%)",
        "Revenue from Churned Customers",
        "Churned Revenue Share (%)",
        "High-Value At-Risk Customers"
    ],
    |
```

```

    "Value": [
        churned_customers,
        churn_rate,
        churned_revenue,
        churned_revenue_percentage,
        high_value_at_risk_count
    ]
})

print("\n===== CHURN INSIGHTS =====\n")

display(churn_insights)

# =====
# SECTION G – PAYMENT INSIGHTS
# =====

# -----
# STEP 31 – MOST USED PAYMENT METHOD
# -----

most_used_payment_row = (
    payment_performance.loc[
        payment_performance["Transactions"].idxmax()
    ]
)
|

```

```
most_used_payment = (  
    most_used_payment_row[  
        "payment_method"  
    ]  
)
```

```
most_used_payment_transactions = (  
    most_used_payment_row[  
        "Transactions"  
    ]  
)
```

```
# -----  
# STEP 32 - HIGHEST REVENUE PAYMENT METHOD  
# -----
```

```
highest_revenue_payment_row = (  
    payment_performance.loc[  
        payment_performance["Revenue"].idxmax()  
    ]  
)
```

```
highest_revenue_payment = (  
    highest_revenue_payment_row[  
        "payment_method"  
    ]  
)
```

```

highest_revenue_payment_value = (
    highest_revenue_payment_row[
        "Revenue"
    ]
)

# -----
# STEP 33 - PAYMENT INSIGHT TABLE
# -----

payment_insights = pd.DataFrame({
    "Insight": [
        "Most Used Payment Method",
        "Highest Revenue Payment Method"
    ],
    "Value": [
        most_used_payment,
        highest_revenue_payment
    ],
    "Metric": [
        most_used_payment_transactions,
        highest_revenue_payment_value
    ]
})

print("\n===== PAYMENT INSIGHTS =====\n")

```

```
display(payment_insights)
```

```
# =====  
# SECTION H – GENERATE EXACTLY 5 RECOMMENDATIONS  
# =====
```

```
# -----  
# STEP 34 – RECOMMENDATION 1  
# Customer retention  
# -----
```

```
recommendation_1 = (  
    f"Launch a targeted retention campaign for "  
    f"high-value at-risk customers. "  
    f"The analysis identified {high_value_at_risk_count:,} "  
    f"customers who have high monetary value but low recent "  
    f"engagement. Use personalized offers, reminders and "  
    f"exclusive benefits to encourage repeat purchases."  
)
```

```
# -----  
# STEP 35 – RECOMMENDATION 2  
# RFM segmentation  
# -----
```

```
recommendation_2 = (
```

```
f"Use RFM-based customer segmentation for personalized "  
f"marketing. Prioritize the '{highest_revenue_segment}' "  
f"segment for loyalty programs and cross-selling, while "  
f"using reactivation campaigns for low-recency segments."
```

```
)
```

```
# -----  
# STEP 36 – RECOMMENDATION 3  
# Product strategy  
# -----
```

```
recommendation_3 = (  
    f"Focus inventory and marketing efforts on the "  
    f"'{best_category}' category because it generates the "  
    f"highest revenue in the analyzed dataset. "  
    f"Combine this category with cross-selling and bundle "  
    f"offers to increase customer order value."  
)
```

```
# -----  
# STEP 37 – RECOMMENDATION 4  
# Purchase timing  
# -----
```

```
recommendation_4 = (  
    f"Schedule promotional campaigns around the strongest "
```



```
f"is {best_purchase_day} at approximately "  
f"{int(best_purchase_hour)}:00. "  
f"Use this period for flash sales, personalized offers "  
f"and campaign reminders."
```

```
)
```

```
# -----  
# STEP 38 – RECOMMENDATION 5  
# Churn reduction  
# -----
```

```
recommendation_5 = (  
    f"Implement an early-warning churn system. "  
    f"The overall churn rate is approximately "  
    f"{churn_rate:.2f}%." "  
    f"Customers showing declining recency and frequency "  
    f"should receive automated retention messages before "  
    f"they become fully inactive."
```

```
)
```

```
# =====  
# STEP 39 – CREATE RECOMMENDATIONS DATAFRAME  
# =====
```

```
recommendations = pd.DataFrame(  
|     "Recommendation_ID": [  
|
```

```
        1,  
        2,  
        3,  
        4,  
        5  
    ],  
    "Priority": [  
        "High",  
        "High",  
        "Medium",  
        "Medium",  
        "High"  
    ],  
    "Recommendation": [  
        recommendation_1,  
        recommendation_2,  
        recommendation_3,  
        recommendation_4,  
        recommendation_5  
    ]  
})
```

```

# -----
# STEP 40 – DISPLAY RECOMMENDATIONS
# -----

print("\n=====")
print("5 ACTIONABLE BUSINESS RECOMMENDATIONS")
print("=====\n")

for _, row in recommendations.iterrows():

    print(
        f"{row['Recommendation_ID']}. "
        f"[{row['Priority']}] "
        f"{row['Recommendation']}\n"
    )

# =====
# SECTION I – CREATE EXECUTIVE SUMMARY
# =====

# -----
# STEP 41 – EXECUTIVE SUMMARY TEXT
# -----

executive_summary = f"""
=====
CUSTOMER BEHAVIOR ANALYSIS – EXECUTIVE SUMMARY
=====

```

BUSINESS PERFORMANCE

Total Customers : {total_customers:,}
Total Transactions : {total_transactions:,}
Total Revenue : {total_revenue:,.2f}
Average Order Value : {average_order_value:,.2f}
Total Quantity Sold : {total_quantity:,}
Return Rate : {return_rate:.2f}%
Churn Rate : {churn_rate:.2f}%

CUSTOMER INSIGHTS

Largest Customer Segment:
{largest_segment}

Highest Revenue Segment:
{highest_revenue_segment}

Top Customer:
{top_customer_id}

Average Customer Spend:
{average_customer_spend:,.2f}

PRODUCT INSIGHTS

Highest Revenue Category:
{best_category}

|

Most Purchased Category:

{most_purchased_category}

Highest Return Rate Category:

{highest_return_category}

PURCHASE BEHAVIOR

Best Purchase Day:

{best_purchase_day}

Peak Purchase Hour:

{int(best_purchase_hour)}:00

Best Time Period:

{best_time_period}

Highest Revenue Month:

{best_month}

PAYMENT BEHAVIOR

Most Used Payment Method:

{most_used_payment}

Highest Revenue Payment Method:

{highest_revenue_payment}

CHURN INSIGHTS

Churned Customers:

```
{churned_customers:,}
```

Churn Rate:

```
{churn_rate:.2f}%
```

Revenue from Churned Customers:

```
{churned_revenue:,.2f}
```

High-Value At-Risk Customers:

```
{high_value_at_risk_count:,}
```

TOP 5 BUSINESS RECOMMENDATIONS

1. Target high-value at-risk customers with personalized retention campaigns.
2. Use RFM segmentation for personalized marketing, loyalty programs and customer reactivation.
3. Prioritize the highest-revenue product category for inventory, cross-selling and promotional campaigns.
4. Schedule promotions around peak customer purchase

5. Implement an early-warning churn system based on recency and purchase frequency.

```
=====
END OF EXECUTIVE SUMMARY
=====
"""

# -----
# STEP 42 - SAVE EXECUTIVE SUMMARY
# -----

with open(
    output_folder / "executive_summary.txt",
    "w",
    encoding="utf-8"
) as file:

    file.write(executive_summary)
```

```
# =====
# SECTION J – SAVE ALL INSIGHTS
# =====

# -----
# STEP 43 – SAVE BUSINESS KPIS
# -----

business_kpis.to_csv(
    output_folder / "business_kpis.csv",
    index=False
)

# -----
# STEP 44 – SAVE PRODUCT INSIGHTS
# -----

product_insights.to_csv(
    output_folder / "product_insights.csv",
    index=False
)

# -----
# STEP 45 – SAVE PURCHASE INSIGHTS
# -----

purchase_insights.to_csv(
```



```
output_folder / "purchase_behavior_insights.csv",  
index=False
```

```
)
```

```
# -----
```

```
# STEP 46 - SAVE CUSTOMER INSIGHTS
```

```
# -----
```

```
customer_insights.to_csv(  
    output_folder / "customer_value_insights.csv",  
    index=False
```

```
)
```

```
# -----
```

```
# STEP 47 - SAVE RFM INSIGHTS
```

```
# -----
```

```
rfm_insights.to_csv(  
    output_folder / "rfm_insights.csv",  
    index=False
```

```
)
```

CHURN INSIGHTS

Churned Customers:

```
{churned_customers:,}
```

Churn Rate:

```
{churn_rate:.2f}%
```

Revenue from Churned Customers:

```
{churned_revenue:,.2f}
```

High-Value At-Risk Customers:

```
{high_value_at_risk_count:,}
```

TOP 5 BUSINESS RECOMMENDATIONS

1. Target high-value at-risk customers with personalized retention campaigns.
2. Use RFM segmentation for personalized marketing, loyalty programs and customer reactivation.
3. Prioritize the highest-revenue product category for inventory, cross-selling and promotional campaigns.
4. Schedule promotions around peak customer purchase

5. Implement an early-warning churn system based on recency and purchase frequency.

```
=====
END OF EXECUTIVE SUMMARY
=====
"""

# -----
# STEP 42 - SAVE EXECUTIVE SUMMARY
# -----

with open(
    output_folder / "executive_summary.txt",
    "w",
    encoding="utf-8"
) as file:

    file.write(executive_summary)
```

```
# =====
# SECTION J – SAVE ALL INSIGHTS
# =====

# -----
# STEP 43 – SAVE BUSINESS KPIS
# -----

business_kpis.to_csv(
    output_folder / "business_kpis.csv",
    index=False
)

# -----
# STEP 44 – SAVE PRODUCT INSIGHTS
# -----

product_insights.to_csv(
    output_folder / "product_insights.csv",
    index=False
)

# -----
# STEP 45 – SAVE PURCHASE INSIGHTS
# -----

purchase_insights.to_csv(
```

```
output_folder / "purchase_behavior_insights.csv",  
index=False
```

```
)
```

```
# -----
```

```
# STEP 46 - SAVE CUSTOMER INSIGHTS
```

```
# -----
```

```
customer_insights.to_csv(  
    output_folder / "customer_value_insights.csv",  
    index=False
```

```
)
```

```
# -----
```

```
# STEP 47 - SAVE RFM INSIGHTS
```

```
# -----
```

```
rfm_insights.to_csv(  
    output_folder / "rfm_insights.csv",  
    index=False
```

```
)
```

```
# -----  
# STEP 48 - SAVE CHURN INSIGHTS  
# -----  
  
churn_insights.to_csv(  
    output_folder / "churn_insights.csv",  
    index=False  
)  
  
# -----  
# STEP 49 - SAVE PAYMENT INSIGHTS  
# -----  
  
payment_insights.to_csv(  
    output_folder / "payment_insights.csv",  
    index=False  
)  
  
# -----  
# STEP 50 - SAVE RECOMMENDATIONS  
# -----  
  
recommendations.to_csv(  
    output_folder / "business_recommendations.csv",  
    index=False  
)
```

```
# =====  
# SECTION K – CREATE MASTER INSIGHTS TABLE  
# =====  
  
# -----  
# STEP 51 – MASTER INSIGHTS  
# -----
```

```
master_insights = pd.DataFrame({  
    "Category": [  
        "Business",  
        "Business",  
        "Customer",  
        "Customer",  
        "Product",  
        "Product",  
        "Purchase Behavior",  
        "Purchase Behavior",  
        "Churn",  
        "Churn",  
        "RFM",  
        "Payment"  
    ],  
    "Insight": [  
        "Total Revenue",  
        "Average Order Value",  
        "Largest Customer Segment",  
        "Top Customer",  
        "Highest Revenue Category",
```

```

        "Highest Revenue Category",
        "Most Purchased Category",
        "Best Purchase Day",
        "Peak Purchase Hour",
        "Churn Rate",
        "High-Value At-Risk Customers",
        "Highest Revenue Segment",
        "Most Used Payment Method"
    ],
    "Value": [
        total_revenue,
        average_order_value,
        largest_segment,
        top_customer_id,
        best_category,
        most_purchased_category,
        best_purchase_day,
        f"{int(best_purchase_hour)}:00",
        churn_rate,
        high_value_at_risk_count,
        highest_revenue_segment,
        most_used_payment
    ]
})

```

```

master_insights.to_csv(
    output_folder / "master_business_insights.csv",
    index=False
)

```



```

)

# =====
# SECTION L - SAVE HIGH-VALUE AT-RISK CUSTOMERS
# =====

if not high_value_at_risk.empty:

    high_value_at_risk.to_csv(
        output_folder / "high_value_at_risk_customers.csv",
        index=False
    )

# =====
# SECTION M - FINAL OUTPUT CHECK
# =====

# -----
# STEP 52 - DISPLAY OUTPUT FILES
# -----

print("\n=====")
print("STEP 8 COMPLETED SUCCESSFULLY")
print("=====\n")

print("Generated Files:\n")

```

```
print("Generated Files:\n")

for file in sorted(output_folder.iterdir()):

    print("✓", file.name)

# -----
# STEP 53 – DISPLAY FINAL RECOMMENDATIONS
# -----

print("\n=====")
print("FINAL 5 BUSINESS RECOMMENDATIONS")
print("=====\\n")

display(recommendations)

# -----
# STEP 54 – DISPLAY EXECUTIVE SUMMARY
# -----

print(executive_summary)

# =====
# END OF STEP 8
# =====
```

===== BUSINESS KPIs =====

	Metric	Value
0	Total Customers	4.967300e+04
1	Total Transactions	2.500000e+05
2	Total Revenue	6.813427e+08
3	Average Order Value	2.725371e+03
4	Total Quantity	7.497240e+05
5	Average Quantity per Transaction	2.998896e+00
6	Total Returns	1.007690e+05
7	Return Rate (%)	4.030760e+01
8	Churned Customers	9.942000e+03
9	Churn Rate (%)	2.001490e+01

===== PRODUCT INSIGHTS =====

	Insight	Value	Metric
0	Highest Revenue Category	Books	2.049396e+08

1	Most Purchased Category	Clothing	7.505200e+04
2	Highest Return Rate Category	Home	4.041000e+01

===== PURCHASE BEHAVIOR INSIGHTS =====

	Insight	Value	Metric
0	Best Purchase Day	Wednesday	36048
1	Peak Purchase Hour	17:00	10652
2	Best Time Period	Night	83032
3	Highest Revenue Month	2020-12	16285542

===== CUSTOMER VALUE INSIGHTS =====

	Insight	Value
0	Top Customer	36437.000000
1	Top Customer Spend	55339.000000
2	Average Customer Spend	13716.559962
3	Average Purchase Frequency	5.032915

===== RFM INSIGHTS =====

	Insight	Value	Metric
0	Largest Customer Segment	Hibernating	9.347000e+03
1	Highest Revenue Segment	Champions	1.893934e+08
2	Highest Churn Segment	Potential Loyalists	2.097590e+01

===== CHURN INSIGHTS =====

	Insight	Value
0	Total Churned Customers	9.942000e+03
1	Overall Churn Rate (%)	2.001490e+01
2	Revenue from Churned Customers	1.360362e+08
3	Churned Revenue Share (%)	1.996589e+01
4	High-Value At-Risk Customers	4.226000e+03

===== PAYMENT INSIGHTS =====

	Insight	Value	Metric
0	Most Used Payment Method	Credit Card	100486
1	Highest Revenue Payment Method	Credit Card	274152396

=====

5 ACTIONABLE BUSINESS RECOMMENDATIONS

=====

1. [High] Launch a targeted retention campaign for high-value at-risk customers. The analysis identified 4,226 customers who have high monetary value but low recent engagement. Use personalized offers, reminders and exclusive benefits to encourage repeat purchases.
2. [High] Use RFM-based customer segmentation for personalized marketing. Prioritize the 'Champions' segment for loyalty programs and cross-selling, while using reactivation campaigns for low-recency segments.
3. [Medium] Focus inventory and marketing efforts on the 'Books' category because it generates the highest revenue in the analyzed dataset. Combine this category with cross-selling and bundle offers to increase customer order value.
4. [Medium] Schedule promotional campaigns around the strongest customer purchase period. The peak transaction period is Wednesday at approximately 17:00. Use this period for flash sales, personalized offers and campaign reminders.
5. [High] Implement an early-warning churn system. The overall churn rate is approximately 20.01%. Customers showing declining recency and frequency should receive automated retention messages before they become fully inactive.

=====

STEP 8 COMPLETED SUCCESSFULLY

=====

Generated Files:

- ✓ business_kpis.csv
- ✓ business_recommendations.csv
- ✓ churn_insights.csv
- ✓ customer_value_insights.csv
- ✓ executive_summary.txt
- ✓ high_value_at_risk_customers.csv
- ✓ master_business_insights.csv
- ✓ payment_insights.csv
- ✓ product_insights.csv
- ✓ purchase_behavior_insights.csv
- ✓ rfm_insights.csv

=====

FINAL 5 BUSINESS RECOMMENDATIONS

=====

Recommendation_ID		Priority	Recommendation
0	1	High	Launch a targeted retention campaign for high-...
1	2	High	Use RFM-based customer segmentation for person...
2	3	Medium	Focus inventory and marketing efforts on the '...
3	4	Medium	Schedule promotional campaigns around the stro...

4

5

High

Implement an early-warning churn system. The o...

=====

CUSTOMER BEHAVIOR ANALYSIS – EXECUTIVE SUMMARY

=====

BUSINESS PERFORMANCE

Total Customers	: 49,673
Total Transactions	: 250,000
Total Revenue	: 681,342,683.00
Average Order Value	: 2,725.37
Total Quantity Sold	: 749,724
Return Rate	: 40.31%
Churn Rate	: 20.01%

CUSTOMER INSIGHTS

Largest Customer Segment:
Hibernating

Highest Revenue Segment:
Champions

Top Customer:
36437

Average Customer Spend:

Top Customer:

36437

Average Customer Spend:

13,716.56

PRODUCT INSIGHTS

Highest Revenue Category:

Books

Most Purchased Category:

Clothing

Highest Return Rate Category:

Home

PURCHASE BEHAVIOR

Best Purchase Day:

Wednesday

Peak Purchase Hour:

17:00

Best Time Period:

Night

Highest Revenue Month:

2020-12

PAYMENT BEHAVIOR

Most Used Payment Method:

Credit Card

Highest Revenue Payment Method:

Credit Card

CHURN INSIGHTS

Churned Customers:

9,942

Churn Rate:

20.01%

Revenue from Churned Customers:

136,036,157.00

High-Value At-Risk Customers:

4,226

TOP 5 BUSINESS RECOMMENDATIONS

1. Target high-value at-risk customers with personalized retention campaigns.
2. Use RFM segmentation for personalized marketing, loyalty programs and customer reactivation.

TOP 5 BUSINESS RECOMMENDATIONS

1. Target high-value at-risk customers with personalized retention campaigns.
2. Use RFM segmentation for personalized marketing, loyalty programs and customer reactivation.
3. Prioritize the highest-revenue product category for inventory, cross-selling and promotional campaigns.
4. Schedule promotions around peak customer purchase periods.
5. Implement an early-warning churn system based on recency and purchase frequency.

=====

END OF EXECUTIVE SUMMARY

=====